

Joins:

1. Inner Join:

```
CREATE TABLE Department (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50)  
);
```

```
CREATE TABLE Employee (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    dept_id INT  
);
```

```
INSERT INTO Department VALUES  
(1, 'HR'),  
(2, 'IT'),  
(3, 'Finance'),  
(4, 'Marketing');
```

2.

```
INSERT INTO Employee VALUES  
(101, 'Rahul', 1),  
(102, 'Sneha', 2),  
(103, 'Vikram', 3),  
(104, 'Anjali', 4);
```

```
SELECT Employee.emp_id,  
       Employee.emp_name,  
       Employee.dept_id,  
       Department.dept_name  
FROM Employee  
INNER JOIN Department  
ON Employee.dept_id = Department.dept_id;
```

3.

Output:

	emp_id	emp_name	dept_id	dept_name
▶	101	Rahul	1	HR
	102	Sneha	2	IT
	103	Vikram	3	Finance
	104	Anjali	4	Marketing

2. Right Join:

```
CREATE TABLE Course (  
    course_id INT PRIMARY KEY,  
    course_name VARCHAR(50)  
);
```

```
CREATE TABLE Student (  
    student_id INT PRIMARY KEY,  
    student_name VARCHAR(50),  
    course_id INT  
);
```

```
INSERT INTO Course VALUES  
(1, 'Java'),  
(2, 'Python'),  
(3, 'Database'),  
(4, 'Networking');
```

```

INSERT INTO Student VALUES
(101, 'Arjun', 1),
(102, 'Priya', 2),
(103, 'Kiran', 1);

SELECT Student.student_id,
       Student.student_name,
       Course.course_id,
       Course.course_name
FROM Student
RIGHT JOIN Course
ON Student.course_id = Course.course_id;

```

Output:

	student_id	student_name	course_id	course_name
►	103	Kiran	1	Java
	101	Arjun	1	Java
	102	Priya	2	Python
	NULL	NULL	3	Database
	NULL	NULL	4	Networking

3. Left Join:

```
CREATE TABLE Customer (  
    customer_id INT PRIMARY KEY,  
    customer_name VARCHAR(50)  
);
```

```
CREATE TABLE Orders (  
    order_id INT PRIMARY KEY,  
    customer_id INT,  
    product VARCHAR(50)  
);
```

```
INSERT INTO Customer VALUES  
(1, 'Arjun'),  
(2, 'Priya'),  
(3, 'Kiran'),  
(4, 'Ananya');
```

4.

```
INSERT INTO Orders VALUES  
(101, 1, 'Laptop'),  
(102, 2, 'Mouse'),  
(103, 1, 'Keyboard');
```

```
SELECT Customer.customer_id,  
       Customer.customer_name,  
       Orders.order_id,  
       Orders.product  
FROM Customer  
LEFT JOIN Orders  
ON Customer.customer_id = Orders.customer_id;
```

5.

Output:

	customer_id	customer_name	order_id	product
▶	1	Arjun	103	Keyboard
	1	Arjun	101	Laptop
	2	Priya	102	Mouse
	3	Kiran	NULL	NULL
	4	Ananya	NULL	NULL

4. Fully – Outer Join:

```
CREATE TABLE Student (
    student_id INT PRIMARY KEY,
    student_name VARCHAR(50),
    club_id INT
);
```

```
CREATE TABLE Club (
    club_id INT PRIMARY KEY,
    club_name VARCHAR(50)
);
```

```
INSERT INTO Student VALUES
(101, 'Arjun', 1),
(102, 'Priya', 2),
(103, 'Kiran', 3),
(104, 'Ananya', NULL);
```

```
INSERT INTO Club VALUES
```

```
(1, 'Coding'),  
(2, 'Music'),  
(3, 'Sports'),  
(4, 'Photography');
```

```
SELECT Student.student_id,  
       Student.student_name,  
       Club.club_id,  
       Club.club_name  
FROM Student  
LEFT JOIN Club  
ON Student.club_id = Club.club_id
```

```
UNION
```

```
SELECT Student.student_id,  
       Student.student_name,  
       Club.club_id,  
       Club.club_name  
FROM Student  
RIGHT JOIN Club  
ON Student.club_id = Club.club_id;
```

Output:

	student_id	student_name	club_id	club_name
▶	101	Arjun	1	Coding
	102	Priya	2	Music
	103	Kiran	3	Sports
	104	Ananya	NULL	NULL
	NULL	NULL	4	Photography

5.Cross Join:

```
CREATE TABLE Student (  
    student_id INT PRIMARY KEY,  
    student_name VARCHAR(50)  
);
```

```
CREATE TABLE Course (  
    course_id INT PRIMARY KEY,  
    course_name VARCHAR(50)  
);
```

```
INSERT INTO Student VALUES  
(101, 'Arjun'),  
(102, 'Priya'),  
(103, 'Kiran');
```

```
INSERT INTO Course VALUES  
(1, 'Java'),  
(2, 'Python');
```

6.

```
SELECT Student.student_id,  
       Student.student_name,  
       Course.course_id,  
       Course.course_name
```

```
FROM Student
```

7.

```
CROSS JOIN Course;
```

Output:

	student_id	student_name	course_id	course_name
▶	101	Arjun	2	Python
	101	Arjun	1	Java
	102	Priya	2	Python
	102	Priya	1	Java
	103	Kiran	2	Python
	103	Kiran	1	Java

6. Self Join:

```

CREATE TABLE Employee (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    manager_id INT
);

INSERT INTO Employee VALUES
(101, 'Arjun', NULL),
(102, 'Priya', 101),
(103, 'Kiran', 101),
(104, 'Ananya', 102);

SELECT
    Employee.emp_name AS Employee,
    Manager.emp_name AS Manager
FROM Employee
LEFT JOIN Employee AS Manager
ON Employee.manager_id = Manager.emp_id;

```

Output:

	Employee	Manager
►	Arjun	NULL
	Priya	Arjun
	Kiran	Arjun
	Ananya	Priya

Queries:

```
CREATE TABLE Student(
    Student_id INT,
    Name VARCHAR(50),
    Marks INT,
    Dept_ID INT
);
```

```
INSERT INTO Student VALUES
(1, 'Shreyas', 90, 102),
(2, 'Vaibhav', 40, 101),
(3, 'Sourav', 80, 102),
(4, 'Rohan', 55, 101);
```

```
SELECT Name, Marks
FROM Student
WHERE MARKS > 75;
```

Output:

	Name	Marks
►	Shreyas	90
	Sourav	80

Subqueries:

```
CREATE TABLE Student(  
    Student_id INT,  
    Name VARCHAR(50),  
    Marks INT,  
    Dept_ID INT  
);  
  
INSERT INTO Student VALUES  
(1, 'Shreyas', 90, 102),  
(2, 'Vaibhav', 40, 101),  
(3, 'Sourav', 80, 102),  
(4, 'Rohan', 55, 101);  
  
SELECT Name, Marks  
FROM Student  
WHERE MARKS > (  
    SELECT AVG(Marks)  
    FROM Student  
);
```

Output:

	Name	Marks
►	Shreyas	90
	Sourav	80